

Plugging into the Future: An Exploration of Electricity Consumption Patterns

TEAM ID	SWTID-2026-7684
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	2 MARKS

3.2 – Solution Requirement

Here is your requirement specifications document compiled into complete, formal, and professional sentences.

Functional Requirements

- **Data Collection:** The system must gather electricity consumption data from available public datasets, connected smart meters, or user-uploaded files. [1]
- **Data Storage:** The system must securely store all collected electricity data to prepare it for processing and deep analysis. [2]
- **Data Preprocessing:** The system must automatically clean the incoming datasets by resolving missing values, removing duplicates, and correcting inconsistencies. [3, 4, 5, 6, 7]
- **Consumption Analysis:** The system must evaluate historical electricity usage records to uncover significant behavioural patterns and long-term trends. [8, 9, 10]
- **Visualization Dashboard:** The system must feature a visual dashboard displaying intuitive charts, graphs, and structured reports for easy interpretation. [11]
- **Prediction Module:** The system must forecast future electricity consumption requirements by utilizing trained machine learning models. [12]
- **User Interaction:** The system must allow users to seamlessly browse custom data insights and dynamically explore dashboard controls.
- **Report Generation:** The system must generate comprehensive summaries and detailed analytical reports based on historical data.

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- **Recommendation System:** The system must provide actionable, data-driven energy-saving suggestions tailored to the user's specific consumption behaviour. [13, 14, 15]
- **Monitoring and Updates:** The system must continuously track real-time consumption data and update metrics automatically. [16, 17]

Non-Functional Requirements

- **Performance:** The system must process high volumes of data and deliver accurate electricity forecasting predictions within a minimal processing window.
- **Accuracy:** The system must guarantee highly reliable analytics and precise forecasting output values.
- **Usability:** The system must provide an accessible, simple, and user-friendly interface that accommodates non-technical individuals.
- **Scalability:** The system must seamlessly scale its architecture to accommodate expanding volumes of electricity ingestion data.
- **Security:** The system must employ strict data protection protocols to secure stored user profiles and information assets.
- **Availability:** The system must maintain continuous uptime to ensure reliable, unrestricted access to dashboards and performance reports.
- **Maintainability:** The system must feature modular code structures that allow engineers to perform swift updates and ongoing improvements.
- **Compatibility:** The system must operate consistently across all modern web browsers and major hardware form factors. [18, 19, 20, 21, 22]

Hardware Requirements

- **Processor:** The host system requires a minimum processing power equivalent to an Intel Core i3 processor or higher.
- **RAM:** The host architecture must dedicate a minimum capacity of 4 GB Random Access Memory (RAM) for active data processing.
- **Storage:** The hosting environment must maintain at least 20 GB of available hard disk space for application files and records.
- **Network:** The infrastructure must possess an active internet connection download.